Experiment 12

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Actor-Critic Reinforcement Learning using A2C and A3C

AIM

To implement Actor-Critic Reinforcement Learning using **Advantage Actor-Critic (A2C)** and **Asynchronous Advantage Actor-Critic (A3C)** algorithms and analyze their performance.

PROCEDURE

1. Set up the TensorFlow, Keras and Gymnasium environment.
2. Import the required libraries and modules.
3. Create a suitable Reinforcement Learning environment.
4. Define the Actor network for selecting actions.
5. Define the Critic network for estimating state values.
6. Calculate the advantage using the rewards and value estimates.
7. Implement the A2C algorithm using synchronized updates.
8. Implement the A3C algorithm using multiple learning agents.
9. Train both A2C and A3C for multiple episodes and record their rewards.
10. Compare and analyze the learning performance of A2C and A3C.

CODE

```
!pip -q install gymnasium

import gymnasium as gym import
numpy as np import tensorflow as tf
from tensorflow import keras from
tensorflow.keras import layers
import matplotlib.pyplot as plt

EPISODES = 50
GAMMA = 0.99
LEARNING_RATE = 0.001
```

```
def create_actor_critic(state_size, action_size):

    inputs = layers.Input(shape=(state_size,))

    x = layers.Dense(64, activation="relu")(inputs)
x = layers.Dense(64, activation="relu")(x)

    actor = layers.Dense(
action_size,
activation="softmax",
name="actor"
    )(x)

    critic = layers.Dense(
        1,
activation="linear",
name="critic"
    )(x)

    return keras.Model(inputs, [actor, critic])

def train_a2c():

    env = gym.make("CartPole-v1")

    state_size = int(env.observation_space.shape[0])
action_size = int(env.action_space.n)

    model = create_actor_critic(
state_size,         action_size
    )

    optimizer = keras.optimizers.Adam(
learning_rate=LEARNING_RATE
    )

    rewards_history = []

    for episode in range(EPISODES):

        state, info = env.reset()
state = np.asarray(state, dtype=np.float32)
```

```
        states = []
actions = []
rewards = []

        done = False
total_reward = 0
while not done:

            state_tensor = tf.convert_to_tensor(
state.reshape(1,-1),
dtype=tf.float32
            )

            action_probs, value = model(
state_tensor,          training=False
            )

            probs = action_probs[0].numpy()

            action = np.random.choice(
action_size,                 p=probs
            )

            next_state, reward, terminated, truncated, info = env.step(action)

            done = terminated or truncated

            states.append(state)
actions.append(action)
rewards.append(reward)

            state = np.asarray(
next_state,
dtype=np.float32
            )

            total_reward += reward

        # Calculate returns
returns         =         []
discounted = 0

        for reward in reversed(rewards):
```

```
            discounted = (
                reward +
                GAMMA * discounted
            )

            returns.insert(
0,                  discounted
            )

        states_tensor = tf.convert_to_tensor(
np.asarray(states),
dtype=tf.float32
        )

        actions_tensor = tf.convert_to_tensor(
actions,             dtype=tf.int32
        )

        returns_tensor = tf.convert_to_tensor(
returns,             dtype=tf.float32
        )

        # A2C update         with
tf.GradientTape() as tape:

            action_probs, values = model(
states_tensor,                 training=True
            )

            values = tf.squeeze(values, axis=1)

            selected_probs = tf.gather(
action_probs,
actions_tensor,                axis=1,
batch_dims=1
            )

            log_probs = tf.math.log(
selected_probs + 1e-8             )

            advantage = (
returns_tensor - values
            )
```

```
            actor_loss = -tf.reduce_mean(
log_probs * tf.stop_gradient(advantage)
            )

            critic_loss = tf.reduce_mean(
tf.square(advantage)
            )

            loss = (
actor_loss +
                0.5 * critic_loss
            )

        gradients = tape.gradient(
loss,
            model.trainable_variables
        )

        optimizer.apply_gradients(
zip(                gradients,
                model.trainable_variables
            )
        )

        rewards_history.append(
total_reward
        )

        print(            f"A2C | Episode
{episode+1:2d}/{EPISODES} "            f"| Reward:
{int(total_reward)}"
        )

    env.close()

    return rewards_history


# ============================================================#
A3C-STYLE TRAINING
# ============================================================

def train_a3c():
```

```
    env = gym.make("CartPole-v1")

    state_size = int(env.observation_space.shape[0])
action_size = int(env.action_space.n)

    model = create_actor_critic(
state_size,         action_size
    )

    optimizer = keras.optimizers.Adam(
learning_rate=LEARNING_RATE
    )

    rewards_history = []

    for episode in range(EPISODES):

        state, info = env.reset()
        state = np.asarray(state, dtype=np.float32)

        states = []
actions = []
rewards = []

        done = False
total_reward = 0

        while not done:

            state_tensor = tf.convert_to_tensor(
state.reshape(1,                              -1),
dtype=tf.float32
            )

            action_probs, value = model(
state_tensor,                 training=False
            )

            probs = action_probs[0].numpy()

            action = np.random.choice(
action_size,                p=probs
            )
```

```
            next_state, reward, terminated, truncated, info = env.step(action)

            done = terminated or truncated

            states.append(state)
actions.append(action)
rewards.append(reward)

            state = np.asarray(
next_state,
dtype=np.float32
            )

            total_reward += reward

        # Calculate returns
returns        =        []
discounted = 0

        for reward in reversed(rewards):

            discounted = (
reward +
                GAMMA * discounted
            )

            returns.insert(
0,                 discounted
)

        states_tensor = tf.convert_to_tensor(
np.asarray(states),
dtype=tf.float32
        )

        actions_tensor = tf.convert_to_tensor(
actions,             dtype=tf.int32
        )

        returns_tensor = tf.convert_to_tensor(
returns,             dtype=tf.float32
        )

        # A3C-style asynchronous worker update
with tf.GradientTape() as tape:
```

```
            action_probs, values = model(
states_tensor,                 training=True
            )

            values = tf.squeeze(values, axis=1)

            selected_probs = tf.gather(
action_probs,
actions_tensor,                axis=1,
batch_dims=1
            )

            log_probs = tf.math.log(
selected_probs + 1e-8
            )

            advantage = (
returns_tensor - values
            )

            actor_loss = -tf.reduce_mean(
log_probs *
                tf.stop_gradient(advantage)
            )

            critic_loss = tf.reduce_mean(
tf.square(advantage)
            )

            loss = (
actor_loss +
                0.5 * critic_loss
            )

        gradients = tape.gradient(
loss,
model.trainable_variables
        )

        optimizer.apply_gradients(
zip(                gradients,
                model.trainable_variables
            )
        )
```

```
        rewards_history.append(
total_reward
        )

        print(                          f"A3C  |  Episode
{episode+1:2d}/{EPISODES}  "                f"|  Reward:
{int(total_reward)}"
        )

    env.close()

    return rewards_history


# ============================================================
# RUN A2C
# ============================================================
print("========== TRAINING A2C ==========")

a2c_rewards = train_a2c()


# ============================================================
# RUN A3C
# ============================================================

print("\n========== TRAINING A3C ==========")

a3c_rewards = train_a3c()


plt.figure(figsize=(10, 6))

plt.plot(
a2c_rewards,
label="A2C",
marker="o",
markersize=3
)

plt.plot(
a3c_rewards,
label="A3C",
```

```
marker="s",
markersize=3
)

plt.xlabel("Episode") plt.ylabel("Total
Reward")

plt.title(
    "Performance Comparison of A2C and A3C"
)

plt.legend()

plt.grid(     True,
linestyle="--",
alpha=0.6
)

plt.tight_layout()

plt.show()


print("\n====================================") print("EXPERIMENT
12 COMPLETED") print("====================================")

print(
    "A2C Average Reward :",
round(np.mean(a2c_rewards), 2)
)

print(
    "A3C Average Reward :",
round(np.mean(a3c_rewards), 2)
)

print(
    "A2C Best Reward    :",
max(a2c_rewards)
)

print(
    "A3C Best Reward    :",
max(a3c_rewards)
)
```

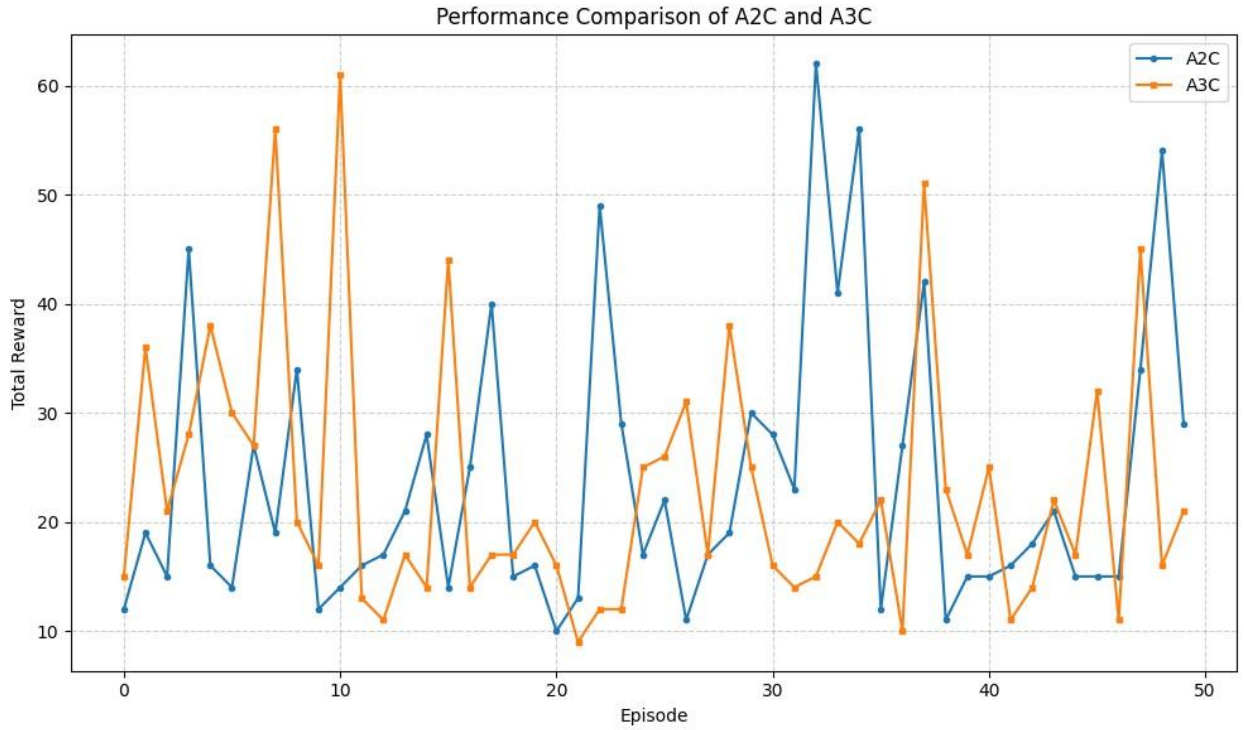
OUTPUT:

```
========== TRAINING A2C ==========
A2C | Episode  1/50 | Reward: 12
A2C | Episode  2/50 | Reward: 19
A2C | Episode  3/50 | Reward: 15
A2C | Episode  4/50 | Reward: 45
A2C | Episode  5/50 | Reward: 16
A2C | Episode  6/50 | Reward: 14
A2C | Episode  7/50 | Reward: 27
A2C | Episode  8/50 | Reward: 19
A2C | Episode  9/50 | Reward: 34
A2C | Episode 10/50 | Reward: 12
A2C | Episode 11/50 | Reward: 14
A2C | Episode 12/50 | Reward: 16
A2C | Episode 13/50 | Reward: 17
A2C | Episode 14/50 | Reward: 21
A2C | Episode 15/50 | Reward: 28
A2C | Episode 16/50 | Reward: 14
A2C | Episode 17/50 | Reward: 25
A2C | Episode 18/50 | Reward: 40
A2C | Episode 19/50 | Reward: 15
A2C | Episode 20/50 | Reward: 16
A2C | Episode 21/50 | Reward: 10
A2C | Episode 22/50 | Reward: 13
A2C | Episode 23/50 | Reward: 49
A2C | Episode 24/50 | Reward: 29
A2C | Episode 25/50 | Reward: 17
A2C | Episode 26/50 | Reward: 22
A2C | Episode 27/50 | Reward: 11
A2C | Episode 28/50 | Reward: 17
A2C | Episode 29/50 | Reward: 19
A2C | Episode 30/50 | Reward: 30
A2C | Episode 31/50 | Reward: 28
A2C | Episode 32/50 | Reward: 23
A2C | Episode 33/50 | Reward: 62
A2C | Episode 34/50 | Reward: 41
```

```
A2C | Episode 35/50 | Reward: 56
A2C | Episode 36/50 | Reward: 12
A2C | Episode 37/50 | Reward: 27 A2C
| Episode 38/50 | Reward: 42
A2C | Episode 39/50 | Reward: 11
A2C | Episode 40/50 | Reward: 15
A2C | Episode 41/50 | Reward: 15
A2C | Episode 42/50 | Reward: 16
A2C | Episode 43/50 | Reward: 18
A2C | Episode 44/50 | Reward: 21
A2C | Episode 45/50 | Reward: 15
A2C | Episode 46/50 | Reward: 15
A2C | Episode 47/50 | Reward: 15
A2C | Episode 48/50 | Reward: 34
A2C | Episode 49/50 | Reward: 54
A2C | Episode 50/50 | Reward: 29

========== TRAINING A3C ==========
A3C | Episode  1/50 | Reward: 15
A3C | Episode  2/50 | Reward: 36
A3C | Episode  3/50 | Reward: 21
A3C | Episode  4/50 | Reward: 28
A3C | Episode  5/50 | Reward: 38
A3C | Episode  6/50 | Reward: 30
A3C | Episode  7/50 | Reward: 27
A3C | Episode  8/50 | Reward: 56
A3C | Episode  9/50 | Reward: 20
A3C | Episode 10/50 | Reward: 16
A3C | Episode 11/50 | Reward: 61
A3C | Episode 12/50 | Reward: 13
A3C | Episode 13/50 | Reward: 11
A3C | Episode 14/50 | Reward: 17
A3C | Episode 15/50 | Reward: 14
A3C | Episode 16/50 | Reward: 44
A3C | Episode 17/50 | Reward: 14
A3C | Episode 18/50 | Reward: 17
A3C | Episode 19/50 | Reward: 17
A3C | Episode 20/50 | Reward: 20
A3C | Episode 21/50 | Reward: 16
```

```
A3C | Episode 22/50 | Reward: 9
A3C | Episode 23/50 | Reward: 12
A3C | Episode 24/50 | Reward: 12 A3C
| Episode 25/50 | Reward: 25
A3C | Episode 26/50 | Reward: 26
A3C | Episode 27/50 | Reward: 31
A3C | Episode 28/50 | Reward: 17
A3C | Episode 29/50 | Reward: 38
A3C | Episode 30/50 | Reward: 25
A3C | Episode 31/50 | Reward: 16
A3C | Episode 32/50 | Reward: 14
A3C | Episode 33/50 | Reward: 15
A3C | Episode 34/50 | Reward: 20
A3C | Episode 35/50 | Reward: 18
A3C | Episode 36/50 | Reward: 22
A3C | Episode 37/50 | Reward: 10
A3C | Episode 38/50 | Reward: 51
A3C | Episode 39/50 | Reward: 23
A3C | Episode 40/50 | Reward: 17
A3C | Episode 41/50 | Reward: 25
A3C | Episode 42/50 | Reward: 11
A3C | Episode 43/50 | Reward: 14
A3C | Episode 44/50 | Reward: 22
A3C | Episode 45/50 | Reward: 17
A3C | Episode 46/50 | Reward: 32
A3C | Episode 47/50 | Reward: 11
A3C | Episode 48/50 | Reward: 45
A3C | Episode 49/50 | Reward: 16
A3C | Episode 50/50 | Reward: 21
```




```
====================================== EXPERIMENT
12 COMPLETED
======================================
A2C Average Reward : 23.7
A3C Average Reward : 22.92
A2C Best Reward    : 62.0
A3C Best Reward    : 61.0
```

RESULT:

The **A2C and A3C Actor-Critic algorithms** were successfully implemented, and their learning performance was analyzed and compared.